AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216A
Cisco Networking (CCCNA) part A

Final Examination - KSA
Second Semester 2025/2026

Date: 10/06/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices. 1 mark per question. **Total: 5 Marks**

1. _____ matches the following scenario: "pretending to be someone else by using stolen personal information to apply for a credit card".
   a. Information theft
   b. Identity theft
   c. Data loss
   d. Disruption of service
2. Which network service automatically assigns IP addresses to devices on the network?
   a. DHCP
   b. Telnet
   c. DNS
   d. Traceroute
3. Which benefit does SSH offer over Telnet for remotely managing a router?
   a. Encryption
   b. Connection via multiple VTY lines
   c. Authorization
   d. TCP usage
4. Which statement is true about Cisco IOS ping indicators?
   a. "!" indicates that the ping was unsuccessful and that the device may have issues finding a DNS server.
   b. "U" may indicate that a router along the path did not contain a route to the destination address and that the ping was unsuccessful.
   c. "." Indicates that the ping was successful but the response time was longer than normal.
   d. A combination of "." And "1" indicates that a router along the path did not have a route to the destination address
5. What is the purpose of the network security authentication function?
   a. To keep track of the actions of a user.
   b. To determine which resources a user can access.
   c. To provide challenge and response questions.
   d. To require users to prove who they are.

Part 2: Answer all the questions. 4 marks per question **Total: 16 Marks**

1. Answer the following questions:
   a) Explain how the switch builds its MAC address table. (1 mark)
   b) There are **Ingress** frames and **Egress** frames; briefly discuss the difference between these two types of frames. (1 mark)
   c) Briefly discuss the cut-through forwarding method. (1 mark)
   d) What happens when the switch receives a frame and the destination address is not located in the MAC address table? (1 mark)
2. Answer the following questions:
   a) Which device breaks the broadcast domain. (1 mark)
   b) When a new switch is bought, the interfaces are assigned to which VLAN? (1 mark)
   c) For which type of traffic should the **Management VLAN** be used? (1 marks)
   d) Mention two benefits of VLANs. (1 mark)
3. Answer the following questions:
   a) A separate VLAN is required for voice traffic because it has many requirements; mention 2 of them. (2 marks)
   b) Mention 2 features of the switch that alleviate (reduce) congestion (2 marks)
4. Answer the following questions:
   a) One of the inter-VLAN routing options is **Legacy Inter-VLAN routing**, briefly describe it and discuss its limitation. (2 marks)

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b) What will be the consequence(s) of not enabling Spanning Tree Protocol (STP) (1 mark).
c) In the STP protocols, on which basis is the root bridge elected? (1 mark)

5. Answer the following questions:
   a) List a benefit of EtherChannel (1 mark)
   b) What is the maximum number of Ethernet links we can have in an EtherChannel? (1 mark)
   c) When a DHCP client boots (or otherwise wants to join a network), it begins a four-step process to obtain a lease; refer to figure 1 and place the four messages in the correct order. (2 marks).
   Messages:
   - DHCP Offer
   - DHCP Request
   - DHCP Acknowledgment
   - DHCP Discover

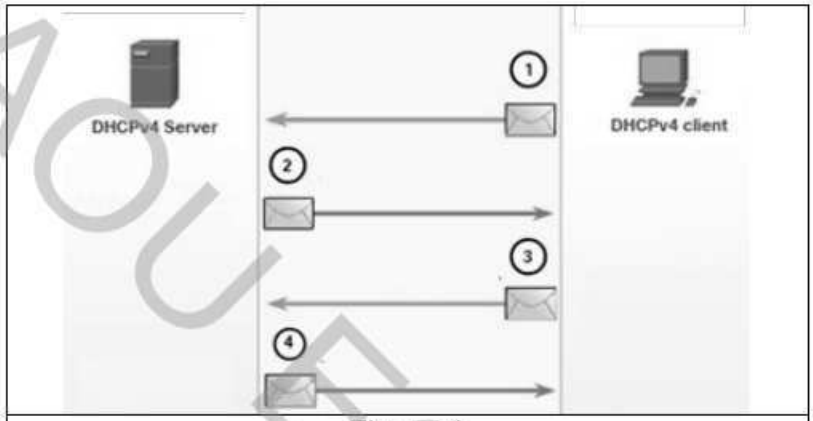


Figure 1

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions related to VLSM (12 marks)
   The following network topology was provided by the network administrator:
   - **LAN1:** has 22 laptops, connected via 3 access points*
   - **LAN2:** has 45 PCs, 45 VoIP phones, 3 network printers, and 7 surveillance cameras connected via 4 switches*.
   - **LAN3:** has 13 servers connected to 3 switches*.
   - **LAN4:** has 14 smart devices and 2 wireless printers connected to 3 access points*.

   * Switch(es) and Access point(s) are accessible remotely (an IP address is needed)

   a) aZSXFCDetermine the number of hosts for each of the 4 networks? (4 marks) *(**Show the steps you followed to obtain the number of hosts**)*
   b) Determine the prefix length for each network (4 marks) *(**Show the steps you followed to obtain the prefix**.)*
   c) Subnet the address block 160.0.15.0/24 as appropriate to provide an efficient use of the addresses but at the same time provide sufficient addresses for all groups and links. Present your answer in a table similar to table1. (4 marks)

Network name	Network address	Host range	Broadcast address	Prefix (or subnet mask)

Table 1

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2. Answer the following questions (9 marks)
   a) Answer the following questions related to STP (6 marks)

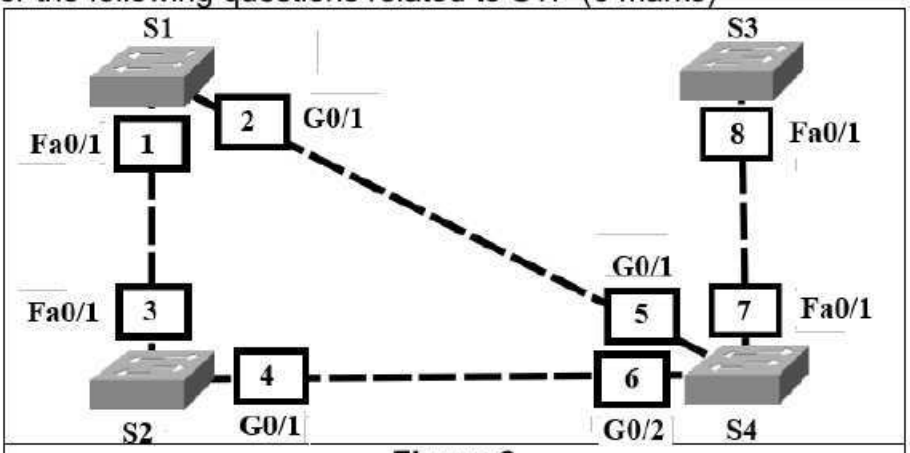


Figure 2

Switch	Priority	MAC address
S1	32769	DC:02:09:F0:00:02
S2	28673	00:00:00:00:00:01
S3	36865	DC:00:00:F0:00:15
S4	24577	EC:02:09:F0:00:01

Table A

Default port costs		
Port	**Link speed**	**Cost (Revised IEEE specification)**
10 Gigabit Ethernet "10G"	10 Gb/s	2
Gigabit Ethernet "G"	1 Gb/s	4
Fast Ethernet "Fa"	100 Mb/s	19
Ethernet "E"	10 Mb/s	100

Table B

a1) Which of the four switches from figure 2 will become the Root Bridge by the STP protocol (Spanning Tree Protocol). Explain the reason (2 marks)
a2) From the answer provided in part a1, determine for each of the 4 switches (as shown in figure 2) the role each port will take. (4 marks)
The role a port can take are:

- Root port
- Designated port
- Alternate port

Note: take into consideration the interface cost (cf. table B).

   b) Using the EUI-64 process and the MAC address DC02:09A7:A002 determine the interface ID of the IPv6 address. Show clearly the steps of calculation. (3 marks)

3. Answer the following questions (8 marks)
   a) Configure on a switch a vlan with the number 52, name it development, and assign three interfaces (Fa0/9, Fa010 and Fa0/11) to this vlan. (2 marks)
   b) The network administrator is configuring interface G0/0 as a trunk link and is setting the native VLAN to 90. Complete the following configuration. (1.5 marks)

```
Interface G0/0
Switchport mode _____
Switchport trunk native _____ _____
```

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c) Complete the router on a stick interVLAN routing configuration for VLAN 77 on interface G0/1. The ip address and subnet mask used for the sub-interface are respectively 192.168.7.5 and 255.255.255.192. (2 marks)

- Interface _____
- _____ dot1Q 77 (0.5 mark)
- _____ 192.168.7.5 255.255.255.192
- Interface _____
- No shutdown

d) The administrator is configuring an EtherChannel between two switches (S1 and S2) using three fast ethernet links. On switch **S2**, the administrator configured the EtherChannel using **channel-group 3** in **PAgP auto mode**. (2.5 marks)

d1) Determine the bandwidth of the EtherChannel (1 mark)

d2) You are tasked with completing the configuration on switch **S1** (1.5 marks)

Switch S1

```
S1(config)#interface range Fa0/1-3
S1(config-if-range)#channel-group 3 mode ______
S1(config-if-range)#exit
S1(config)#interface _____ 3
S1(config-if)#switchport _____ trunk
S1(config-if)#switchport nonegotiate
```

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End of questions